

Voice of the Unheard: Conversational Challenges Between Signers and Non-Signers and Design Interventions for Adaptive SLT Systems

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Abstract

Communication barriers remain a significant challenge for the D/deaf community, affecting social interactions, access to services, and overall quality of life. Despite advancements in sign language technology, existing solutions fall short of addressing the unique needs of the D/deaf community. We explore these needs and challenges by conducting a survey with 200 signers and non-signers. Our findings reveal that while the D/deaf community utilizes technology for communication, obstacles including limited translation accuracy, higher costs, and a lack of learning options persist. Additionally, varied proficiency levels, regional dialects, and the lack of visual cues further complicate interactions. We identified key barriers and design approaches that emphasize inclusivity, adaptability, and usability. By incorporating feedback from the D/deaf community, our work provides insights for developing user-centered sign language systems. Our study highlights the importance of bridging the gap between technological innovation and real-world application, paving the way for accessible and transformative communication tools.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Accessibility**.

Keywords

Sign Language Translation, Accessibility, D/deaf community, Assistive technology, Inclusive design

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1 Introduction

In 2019, 1.57 billion people globally were living with hearing disabilities [16], a number projected to rise to 2.45 billion by 2050 [16]. Alarming, hearing loss is increasing among younger populations [31], with approximately 2 to 3 out of every 1,000 children in the U.S. born with hearing loss in one or both ears [9]. Individuals with hearing loss face setbacks spanning from inadequate learning opportunities [19] to significant employment barriers [9]. Communication barriers further exacerbate these difficulties, limiting social interactions, access to services, and overall quality of life [9]. For approximately 466 million D/deaf individuals worldwide, sign language [1] is the primary mode of communication [10, 25, 29]. However, issues such as lack of awareness [9], different levels of proficiency [27], and accessibility obstacles often impede effective communication [3], highlighting the urgent need for innovative solutions such as wholistic Sign Language Technology (SLT) systems [14].

While significant strides have been made in SLT research such as advancements in transformers [6, 10, 34], computer vision [39], neural networks [39], multi-modal translation approaches [20], and wearable technologies [2, 15, 24, 41], existing systems often lack the comprehensiveness required for widespread adoption [11, 14, 36]. Obstacles such as limited datasets [14, 38], variability in sign languages [36], and real-time processing constraints [36] continue to persist. As such, a gap emerges between the development of SLT systems and their effective use by the D/deaf community [1, 35]. Furthermore, the challenges of communicating with the D/deaf community from the perspectives of people not in the same community remain under-explored.

To address these constraints, we conducted a cross-sectional survey [33] with both signers and non-signers to explore their unique conversational barriers and identify user-focused design requirements. Our findings reveal that while the D/deaf community [1, 35] utilizes technology to aid communication, existing tools are often constrained by limited translation capabilities, higher costs, insufficient learning options, inaccuracy, and lack of versatility and feedback. We also found that signers and non-signers face difficulties in establishing basic communication, even with technological assistance. Alternate communication methods, such as lip reading, hand gestures, and hearing aids, provide some support but are hindered by issues like insufficient resources, background noise, mumbling, lack of eye contact, and limited awareness. Additionally, dialectal and regional variations and signing speeds, contribute to

frustration, slowed communication, physical strain, and misinterpretations.

By highlighting the importance of designing and developing comprehensive SLT systems, we are aligning with the broader goal of advancing human-computer interaction research and adoption in real-world settings [23]. Our study documents the willingness and impediments faced by the D/deaf community in adopting accessible systems and highlights insights for designing user-centered solutions. By engaging the D/deaf community as active contributors, we advocate for systems that prioritize inclusivity, adaptability, and real-world usability. This work presents a significant opportunity to bridge communication gaps and redefine the role of assistive technologies in promoting accessibility.

2 Methodology

We conducted a cross-sectional survey in August 2024 to explore these barriers in conversations between signers and non-signers and to investigate potential technological interventions to address these challenges. The cross-sectional design was selected for its ability to capture diverse perspectives through open-ended questions [8, 33].

2.1 Participants

We recruited participants across two groups: signers and non-signers. Signers were required to have some proficiency in any sign language, while participation in the non-signers group was open to all who self-reported them as non-signers. We used a screening questionnaire to identify individuals with disabilities or hearing loss and their proficiency in using sign language. A total of 200 participants completed the survey, with 100 participants (S1 - S100) in the signer group and another 100 participants in the non-signer group (NS1 - NS100). The participants were recruited through Prolific [26].

In the signer group, 46 participants used she/her pronouns, 52 used he/him pronouns, and two used they/them pronouns. 55 signers identified themselves as beginners in using sign language, 24 signers were novices, 16 signers were proficient, and 5 signers considered themselves experts. The average age of signers was 38.64 ± 11.53 years with 12 signers between ages 18-25, 33 signers between ages 26 - 35, 23 signers between ages 36 - 45, 25 signers between ages 46 - 55, and 7 signers between ages 56-65. 67 signers mentioned using some technology to communicate with others.

In the non-signers group, 52 participants used she/her pronouns, 46 used he/him pronouns, and two used they/them pronouns. 78 non-signers reported having some level of interaction with signers. The average age of non-signers was 33.26 ± 11.06 years with 29 non-signers between ages 18-25, 31 non-signers between ages 26 - 35, 24 non-signers between ages 36 - 45, 14 non-signers between ages 46 - 55, and 2 non-signers between ages 56-65.

2.2 Survey Design

The survey was designed to address three major goals: (1) to understand the conversational constraints between and among signers and non-signers, (2) to explore how current sign language tools address these challenges and (3) to explore alternative technologies to overcome the shortcomings of existing tools. We asked both

groups questions about their hurdles, exposures to technology, experiences, and strategies when interacting with the D/deaf community. Specifically, the questions explored the setbacks signers encounter during their conversations with other signers, the difficulties non-signers face when communicating with signers, and the methods of communication they rely on. We also investigated whether participants use alternative methods of communication beyond sign language and assessed the technologies currently employed by both groups in their interactions. The survey was conducted through Qualtrics [30], an online survey platform, and distributed through Prolific [26]. Each participant was paid \$14 USD per hour. The study was approved by Louisiana State University's institutional review board (IRB).

2.3 Data Analysis

We performed thematic analysis using an inductive process (bottom-up approach) [5] to analyze open-ended survey responses. The first author open-coded all responses to create a codebook. The research team then discussed and finalized the codebook across three sessions. We used Figjam [13] to collect all the codes to create visual mappings based on the responses from the participants. The research group then iteratively identified and grouped the codes into themes across six sessions. The themes, their descriptions, and example codes are presented in Table 1.

3 Survey Findings

Our survey analysis uncovered key insights into the conversational hurdles in sign language communication and proposed design interventions to address these issues. The findings revealed that communication barriers affect both signers and non-signers. While signers are often tech-enabled, they face significant limitations with current technologies. Participants also suggested that SLT systems could extend beyond translation to incorporate inclusive design features.

3.1 Two Worlds, One Language: Barriers in Sign Language Interactions

Signers and non-signers acknowledged practical roadblocks in using sign language during their conversations. Some of these challenges arise from a lack of awareness and understanding. In addition, factors such as signing speed, proficiency levels, and external influences contribute to these barriers. The responses emphasize how these constraints affect both signers and non-signers and are often shaped by underlying unseen factors including psychological challenges.

3.1.1 Challenges during Conversations between Signers and Non-Signers. We found that conversational barriers between signers and non-signers persist throughout the conversations. Both groups frequently needed to explicitly convey their disability and preferred communication methods. For instance, S82, a signer shared: *"Letting them [non-signers] know I cannot hear is a struggle"*. and NS62, a non-signer mentioned: *"They [signers] look normal and so people don't think they may have a disability"*. Participants noted these communication gaps often resulted in misunderstandings. S70 stated: *"I struggle with intent to simply get my point across [to non-signers]"*. On the other side, NS81 emphasized, *"when one does not know*

Table 1: This table represents themes, codes and descriptions identified based on the thematic analysis on the cross-sectional survey.

Themes	Sample Codes	Description
Barriers in Sign Language	Awareness, Communication Speed, Proficiency Levels, Styles, Physical Challenges, Psychological Challenges, External Factors	Hurdles based on different dialects, proficiency and fluency variations, signer status and unseen obstacles
Tech-Exposure of D/deaf Community	Alternate Communication Methods, Interaction Technology, Translation Technology, Current State Limitations	Different types of known technologies, their limitations, available alternative communication methods
Community Focused Design Ideas	Translation Accuracy, Tooling Choice, Feedback Options, Resource Limitations	Importance of accuracy measurement, feedback options, visual cues to support people with disability
Enablement Beyond Sign Transition	Sign Language Learning Options, Accessibility Options, Automated Sign Construction	Options beyond sign translation

sign language, it is extremely hard to interpret what the signs [from signers] actually mean”. Furthermore, the participants indicated that this misunderstanding increased during conversations about complex topics and in crowded environments. For example, S87 expressed: “Explaining complex or abstract concepts in sign language to non-signers can be challenging... Many non-signers may have limited awareness or knowledge about sign language and Deaf culture”. Additionally, Environmental factors, such as noise or distractions, also played a role. For example, NS26 explained: “It is challenging to get the point across when ordering food or speaking on customer service related issues [due to crosstalk]”.

3.1.2 Challenges Within the Signing World. Communication constraints are not isolated to conversations with non-signers, they also exist within the signer community. Our participants highlighted dialects, signing styles, proficiency, and signing speed as significant factors that affect communication. For instance, S23 stated: “Since I am a beginner, I still get words and signs wrong”. S18 added: “Not knowing certain signs leads to my confusion”. Similarly, S74 noted: “My vocabulary is very limited, so I rely on hand gestures”, while S98 emphasized: “I am not proficient with every word, and fast signing makes understanding difficult”.

The participants also stated that speed variations impact conversations. S33 said: “Communication is slow and laborious due to speed differences”, and S52 echoed: “Speed variations always slow [down] communication”. S41 elaborated: “New signers often use finger-spelling, while experienced signers use word signs, but the latter’s speed often makes understanding [it] difficult”. Finger-spelling, while common among new signers, was noted to hinder communication efficiency. S41 commented: “They [new signers] try to finger-spell everything, which takes too long and is not always accurate”.

3.1.3 Exploring Unseen Obstacles in Sign Language Interactions. While variations in fluency and proficiency are significant barriers, the survey also revealed deeper, often overlooked challenges in sign language interactions, including physical struggles like muscle cramping and psychological issues such as frustration, low self-esteem, prejudice, and projection. For instance, S63 expressed: “My fingers cramp up when communicating with someone for a long time”

and S95 wanted more time to communicate: “I take time to understand because I am a bit slow with learning (also autistic)”.

Both signers and non-signers agreed that impatience on both sides often obstructs their conversations. NS97 explained how “the frustration kicks in as signers do not understand”. In contrast, S56 explained frustration from the signer’s perspective by stating, “... just still being a beginner, I don’t want them to feel that I am not trying. I don’t want them [expert signers] to get frustrated”. Participants also noted that there are assumptions and prejudices between novice signers and expert signers. To illustrate, S8 mentioned “Expert signers are not receptive and are judgmental. They can’t understand what I’m trying to say”, and S52 revealed that “...they [expert signers] assume I am already more fluent than I am, leading to frustration in the least and all the way to outright questioning if I’m actually deaf!”.

The participants also conveyed that there are additional hurdles due to the presence of different dialects in sign languages and variations in teaching styles. For instance, S45 mentioned that “there are so many different ways of teaching it [sign language] and there is no standard at this time”, leading to confusion. Furthermore, S89 mentioned: “People use a lot of slang or abbreviations that I do not know” contributing to miscommunication. Similarly, NS92 pointed out about regional variations: “Nuances and special signs used by region are a big factor. I know my New York D/deaf friends sign a lot differently than some of the D/deaf community on the West Coast”.

In addition to variations in sign language, the signers expressed that conversations are interrupted by environmental issues. For instance, S33 highlighted background noises as a distraction and NS13 noted insufficient light as an interruption, especially at night. Both signers and non-signers agreed that lack of eye contact is also an added distraction. NS72 asserted that “turning the head away when talking” is a distraction and S98 agreed that “Sometimes the other person does not look at me when speaking”, referring to the need to maintain the eye contact throughout signed conversations.

3.2 From Signs to Screens: The Signers are Tech-Ready

Participants reported using various alternative communication methods beyond sign language, including hand gestures, hearing aids, lip reading, and writing, often tailored to specific environments. For instance, S100 shared: *“I can’t use sign language, and I need alternate forms of communication like writing on paper and gesturing”*. While some of these methods were enhanced by technology, participants highlighted issues such as additional costs and limited accessibility.

Participants frequently relied on interaction technology devices such as text messages, note-taking applications, emails, and notepads to convey their intent. Text messaging, in particular, was valued for its ubiquity. S82 noted: *“Text messages are pretty universal”*, while S27 shared: *“It is easy for non-signers to understand text messages”*. S41 added: *“I can type exactly what I want to say and show someone right away [without delay]”*. Translation technologies were also occasionally used, including tools like Bloom [4], Glove[28], Lingvano [22], various Text-to-Speech applications, and Video Relay Service (VRS)[12]. S71 expressed an appreciation for translator application: *“Intuitive controls in translator apps and a simple interface make it easier for me to navigate and use the application effectively. It also allows me to say it out loud by interpreting what I want to convey”*.

Despite their reliance on technology and alternative communication methods, participants highlighted some notable limitations. For instance, dependency on devices was a recurring issue. S1 explained: *“I need to have my phone with me at all times”*, and S6 noted: *“It is difficult when there is no pen or paper around for me to write what I want to say”*. Participants also encountered setbacks specific to certain methods. Lip reading, for example, was impacted by unclear articulation. S60 mentioned: *“If they mumble and are slow, it is hard to read their lips”*. Hearing aids and live captions also had limitations. S57 pointed out: *“Hearing aids and live captions can’t fix everything, especially background noise that makes understanding difficult. Live captions aren’t always accurate and often have delays, which disrupt the flow of conversation”*. Cost barriers were another significant issue. S85 emphasized: *“I cannot get new ones [hearing aids] because of insurance changes and limited coverage options”*. Our participants provided valuable insights into addressing these gaps through the development of inclusive and adaptable Sign Language Translation (SLT) systems.

3.3 SLTs Should Account for Versatility, Accuracy, Privacy, and Feedback for Comprehensive Translation

Our participants emphasized the need for comprehensive SLT systems that can incorporate a range of characteristics and demonstrate rigorous flexibility to address diverse needs of the participants. S57 remarked: *“ASL translation is notoriously unreliable. While it can generally read/understand basic finger spelling and signs, so far I have yet to see a system that can understand ASL grammar and keep up with the speed of native signers, while also understanding people (like myself) who aren’t as fluent”*. These responses highlight the importance of addressing both functional and emotional aspects of communication. Based on participants’ feedback, we identified

four key factors that could enhance the comprehensiveness and effectiveness of SLT systems.

3.3.1 Translation Versatility is a Key Differentiator in SLTs. Participants highlighted that while current SLTs reliably support simple conversations, they often fail in complex settings due to context, dialects, and regional variations. For instance, S86 noted: *“Just like spoken languages, sign languages can vary in terms of regional accents or dialects, leading to misunderstandings”*. Key improvements suggested by participants included incorporating contextual understanding, facial expressions, vocal intonations, and inclusivity for skin tone variations. NS40 emphasized: *“AI-enabled systems should understand and adjust translations based on context and nuances”*, while NS85 added: *“Let users pick ASL styles, ensure it gets the context right, and include visual tools for better comprehension”*.

3.3.2 Enhancing Clarity Through Translation Accuracy and Feedback Mechanisms. Participants stressed that translation accuracy and feedback mechanisms are critical for SLTs. S86 highlighted the need for high accuracy in recognizing signs to avoid errors, while S100 emphasized the importance of translating signs into coherent sentences rather than exact signed words. Feedback mechanisms were seen as essential for continuous improvement. NS83 suggested enabling users to correct inaccuracies, and NS98 noted: *“A system capable of adapting to user feedback would improve over time”*.

3.3.3 Translation Systems Should Allow Personalization Preferences. Participants advocated for personalization features to address varying proficiency levels and individual preferences. Adjustable signing speed was a recurring request, with S60 and S36 suggesting this feature for beginners. S52 proposed adding the flexibility to switch between finger spelling and signed words. Personalization also extended to saving conversation history and creating user-specific reference lists. NS89 emphasized: *“Saving and recording signed conversations for future reference would be helpful”*, while S8 expressed a desire for systems that *“translate sign language using my own voice”*.

3.3.4 Enhancing Inclusivity: Integrating Auditory Support and Visual Cues in SLTs. Participants underscored the importance of integrating auditory and visual support into SLT systems to improve accessibility. Visual signals, such as captions, were deemed crucial for verifying content, with S76 stating: *“Seeing the signs and captions visually helps a deaf person like me”*. Audio features were also suggested to be potentially beneficial in maintaining eye contact and enabling smoother group conversations. S85 shared: *“Audio support would allow me to hear the translation while maintaining eye contact”*. These features were seen as vital for making SLTs more inclusive and effective for diverse users.

3.4 Need for End-to-End Translation Systems that Unifies Learning, Resource Sharing, and Sign Creation

Participants from both groups emphasized the need for SLTs to evolve into comprehensive communication support systems by integrating alternative technologies and going beyond simple sign language translation. These enhancements include automated sign

construction, access to common language resources, and options for learning and practicing sign language within the same platform.

3.4.1 Automated Sign Construction. Non-signers, in particular, expressed a preference for automated sign construction as a way to enhance communication. NS81 suggested: “...the ability to turn spoken language into images of sign language”, while NS8 highlighted the benefit of “automatically constructing sign language [images] as we type”. Signers also saw potential in automated tools, as S39 mentioned the usefulness of “...phrase suggestions when responding to communication partners”. These responses underline the value of multi-modal translation across spoken and text-based SLT systems, particularly for simplifying signed conversations. However, participants stressed the importance of ensuring these systems remain intuitive and non-distracting. NS28 noted: “SLTs need to be simple enough and not a distraction that deters people from having a genuine conversation”.

3.4.2 Incorporating Emojis and Pre-loaded Phrases. While non-signers leaned toward automated sign construction, signers wanted sign language-based emojis to convey emotions. S34 remarked: “...I would like to see emojis for deciphering and communicating emotions”. Additionally, S70 proposed preloaded generic responses to expedite conversations. Context-specific phrases also emerged as a priority. NS24 suggested: “A system with a list of common phrases for each environment (grocery, healthcare, fast food restaurants, etc.) that can easily be clicked and shown to another person would be helpful [for conversations]”. These additions could enhance usability and streamline interactions in diverse settings.

3.4.3 Learning and Practicing Sign Language: Participants from both groups expressed interest in using SLTs as learning tools to practice sign language. Non-signers advocated for features such as “study sessions” (NS59) and a “learning mode” (NS28), while beginner-level signers emphasized the need for practice tools. S58 suggested a “quiz mode that will show pictures of the sign and test” users. Similarly, S24 wanted a feature to “provide signs for me to sign back so that I can practice those signs”. The need for accessible learning options was further echoed by NS1, who emphasized the importance of reducing dependency on SLT systems: “The app [SLT system] should enable e-learning so that the hearing community can learn the signs, not just interpret them”.

4 Discussion

4.1 Breaking Barriers by Improving Confidence and Connection in Signed Conversations

Our findings in this work suggest that signers often feel challenged when disclosing their disability to both non-signers as well as other signers. Non-signers often lack awareness and struggle to recognize the necessity of sign language for effective communication. Expert signers often become frustrated due to a lack of proficiency in new signers, raising questions about whether they belong in the D/deaf communities. Such difficulties in assessing proficiency levels during signed conversations exacerbate frustrations, discomfort, and misplaced assumptions, ultimately contributing to diminished confidence within the D/deaf community. These findings corroborate with prior works [40] that highlight that individuals in the

D/deaf community often experience anxiety and discomfort during communication with others.

To overcome these impediments, future SLTs could integrate options to declare people’s communication profiles as part of the introductory handshake. Then, visual cues, such as color-coded badges or proficiency indicators, could then communicate signing status, skill levels, and other profile information before the conversation begins [18, 37]. Other approaches to reduce frustration during a signed conversation could integrate smart feedback that tracks conversation flow [14] and notifies a signer if they are going too fast or too slow which may disrupt the conversation. Furthermore, researchers could also explore AI-assisted agents as conversation mediators that will facilitate signed conversations across all parties. Such approaches have seen success in spoken and written conversations in both meetings and professional domains [21]. However, due to the unique nature of signed conversations, further investigation is required to assess the effectiveness of such approaches.

4.2 Enhancing SLT Systems: Unity in Diversity in Sign Language Communication

Our participants reported struggling to sustain conversations due to sign language variations including dialect and regional differences, which lead to inaccuracies in translation in SLTs. These findings align with prior research [11] that suggests how translation inaccuracies persist in current SLT systems. They also emphasize that various sign language methods including semantic machine translation, statistical machine translation, hardware-based approaches, and software-based approaches often result in standalone SLT systems that lack integration and interoperability.

One avenue to overcome regional variations is to build computational approaches that can account for such variations. Especially, in machine learning-based systems, robust datasets are crucial to building accurate SLT systems [10, 38]. To that end, research efforts are needed to collect and consolidate robust datasets to train better models. More holistically, researchers could explore frameworks that go beyond translation and enable users to customize regional preferences, preferred dialects, signing styles, and even domain-specific contexts to enhance translation accuracy [7, 32]. Participants further emphasized the importance of incorporating dynamic feedback mechanisms within SLT systems to improve accuracy, offer real-time corrections, and facilitate a smoother and more seamless translation experience. Such frameworks could expand beyond regional differences to entirely different sign languages. While over 130 sign languages exist globally [17], current SLT systems predominantly support specific languages such as ASL, JSL, BSL, LSF, CSL, and AUSLAN [38]. In the future, SLT systems could adopt a unified framework by integrating multiple sign languages, enabling seamless communication across linguistic boundaries. These systems could support more than one sign language within a single conversation, paving the way for truly inclusive and adaptive communication solutions.

4.3 Integrating Learning, Translation, and Accessibility to Create Adaptive SLT Systems

Based on participants' feedback, we found that many non-signers lack adequate literacy in sign language, despite interacting with members of the D/deaf community in daily conversations. Both signers and non-signers agreed that this lack of sign language literacy reduces their social interaction. This finding aligns with prior work [19], which highlights how limited awareness marginalizes the D/deaf community. For example, research on sign language learning [10] and translation [2, 6, 11, 34] illustrates progress in isolated domains.

In the future, SLT systems could help learn how to sign by incorporating sign constructions and interactive lessons. This could include offline learning opportunities as well as playback options to enable users to revisit conversations and practice sign language, fostering self-paced, offline skill development. Moreover, to raise awareness about the D/deaf community, SLT systems could be deployed and integrated in critical environments, such as health-care facilities, retail spaces, and eateries. In many such environments physical accommodations for the D/deaf community remain scarce [19]. Such environments could incorporate environment-specific accessible solutions, such as sign translation kiosks for ordering food or communicating medical concerns in clinical settings. These approaches could offer two distinct benefits: (1) providing support for the D/deaf community and (2) normalizing accessible technologies for the D/deaf community to make others aware of their presence in communal places.

5 Conclusion

Despite the advances in sign language translation (SLT) systems, communication between signers and non-signers remains challenging due to a lack of comprehensive solutions. To learn more about these challenges and explore potential solutions, we surveyed 200 participants. Our participants' responses highlight the challenges signers and non-signers face in their communication and emphasizes the need for inclusive and adaptable Sign Language Technology (SLT) systems. Participants identified automated sign construction, context-aware translations, learning tools, and inclusivity as key design considerations for a comprehensive SLT system. These insights underscore the importance of user-centered, versatile, and accessible solutions. By addressing these gaps, SLT systems can foster seamless communication between the D/deaf and hearing communities, paving the way for a more inclusive and connected world.

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